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1. Aerenchyma:
Parenchyma which contains air in its intercellular spaces. It helps in aeration and buoyancy. Example: *Nymphae and Hydrilla*.

5. Prosenchyma:
Parenchyma cells became elongated, pointed and slightly thick walled. It provides mechanical support.

Parenchyma

2. Storage Parenchyma:
Parenchyma stores food materials. Example: Root and stem tubers.

4. Chlorenchyma
Parenchyma cells with chlorophyll. Function is photosynthesis. Example: Mesophyll of leaves.

3. How?.... Stellate Parenchyma
Star shaped parenchyma. Example: Petioles of *Banana and Canna*.

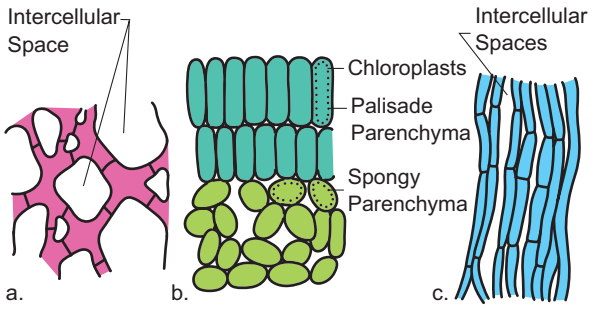


Figure 9.5: c) Stellate parenchyma, d) Chlorenchyma, e) Prosenchyma

Collenchyma (Gk. Colla-glue; enchyma – an infusion)

Collenchyma is a simple, living mechanical tissue. Collenchyma generally occurs in hypodermis of dicot stem. It is absent in the roots and also occurs in petioles and pedicels. The cells are elongated and appear polygonal in cross section. The cell wall is unevenly thickened. It contains more of hemicellulose and pectin besides cellulose. It provides mechanical support and elasticity to the growing parts of the plant. Collenchyma consists of narrow cells. It has only a few small chloroplast or none. Tannin maybe present in collenchyma.Based on pattern of pectinisation of the cell wall, there are three types of collenchyma

Types of Collenchyma

1. Angular collenchyma
It is the most common type of collenchyma with irregular arrangement and thickening at the angles where cells meets. Example:Hypodermis of *Datura* and *Nicotiana*

2. Lacunar collenchyma
The collenchyma cells are irregularly arranged. Cell wall is thickening on the walls bordering intercellular spaces. Example:Hypodermis of *Ipomoea*

3. Lamellar collenchyma
The collenchyma cells are arranged compactly in layers(rows). The Cell wall is thickening is at tangential walls.These thickening appear as successsive tangential layers. Example:Hypodermis of *Helianthus*